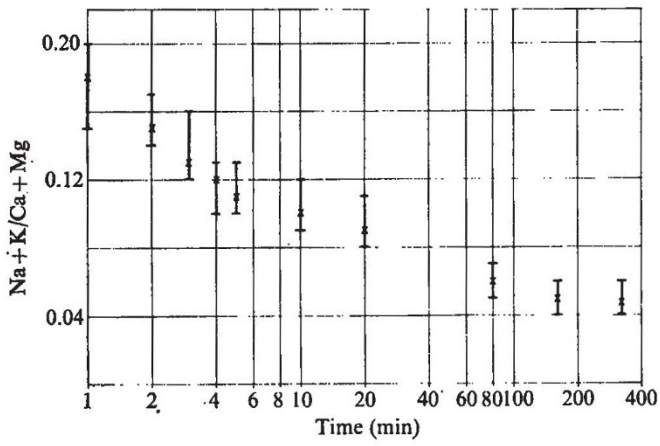


Fig. 2 Ratio of alkalis to alkaline earths as a function of time in the experiment. Time in this figure is the time elapsed between the contact of the first drop with the particles to the completed filtering.

If we now consider the mechanism discussed above, a flushing-out of ions along with the squeezed water is likely to occur. Due to a more rapid migration of alkalis which have larger diffusion coefficients than Ca or Mg, the squeezed water can be selectively enriched in Na and K. On the other hand, forcing of squeezed water across a frozen mud layer at the base can also produce an increase in Na+K/Ca+Mg ratio; indeed such a layer may be considered as an ion-exchanger membrane causing electrolyte filtration[6]. Thus the peculiar chemical characteristics of re-gelation ice are explained by refreezing of this squeezed water at the ice-bedrock interface. This gives some weight to the assertion that pressure-melting within the basal ice accompanied by meltwater squeezing and re-gelation is an active process at the base of an alpine glacier. The considerations developed above point to the need for a re-appraisal of the commonly used methods for calculating basal sliding of glaciers over surfaces of varying roughnesses.

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Evidence from Iceland on geomagnetic reversal during the Wisconsinan Ice Age

The discoveries of reversely magnetised basalts in the axial valley of the Mid-Atlantic Ridge[1–3] and Cayman Trough[4] have added a new facet to the debate about geomagnetic events during the recent Brunhes. Are these reversely magnetised rocks the legacy of a recent geomagnetic event or are they the product of some process of axial valley tectonics which we do not yet understand? We have confirmed the existence of two reversely magnetised lava flows which were erupted subglacially in southwestern Iceland (Fig. 1) and seem to be direct analogues of the reversed rocks in the axial valley. We report here that a recent geomagnetic event is the only mechanism which can explain both the Icelandic and axial valley reversed outcrops.

The reversely magnetised Icelandic material, which was originally reported by Einarsson[5] on the basis of measurement of partially orientated hand samples, forms the top of the small subglacial ridge near Maelifell, south of Thingvallavatn. It is poorly exposed in scree and outcrops in spheroidal, pillow-like structures of olivine rich basalt.

We sampled the 11 outcrops which seemed to be in place. Most inclinations steepened during cleaning in alternating fields (Fig. 2). The natural remanent intensities were unusually strong and ranged over 5–192 A m^{-1} with an average of 50 A m^{-1}. The median destructive fields were unusually low for young subaqueous lavas; they ranged from 4 to 25×10^3 A m^{-1} and averaged 9.6×10^3 A m^{-1}. Curie temperatures were run for seven specimens, and they ranged from 222 to 272 °C, with an average of 248 °C. There was no evidence of multiple Curie temperatures in any sample.

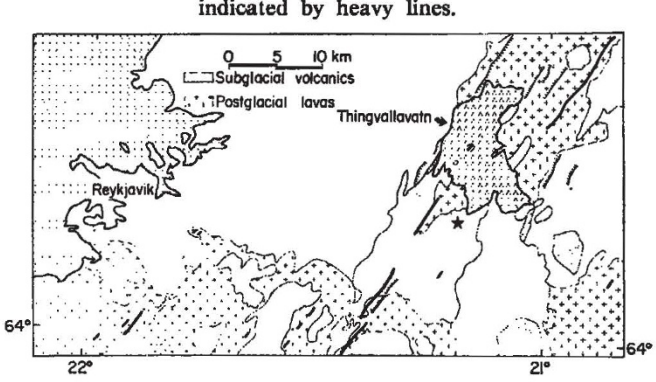
Cleaned directions, averaged by individual outcrop, are shown in Fig. 3. It is evident that there are two distinct directions defined by the data and a third direction is defined by only one outcrop. We interpret these directions as evidence that two separate lava cooling units were extruded during a time of reversed magnetic polarity. Sufficient time must have passed between the eruption of the first unit and that of the second unit to allow a large amount of secular variation to occur. The distribution of samples on the hillside indicates that unit *a* is the youngest.

Samples from pillows below the reversed units define a normal direction of $I=+62°$, $D=331°$. The solitary shallow direction lies between the normal and reversed units and it may represent a transition direction or simply mean that the outcrop was not in place.

The reversed outcrop occurs near the top of the stratigraphic sequence of subglacial volcanics which form the Hengill complex. These volcanics are thought to have been formed under ice of Wisconsinan age[6]. Unfortunately, there are no radiometric dates to confirm this interpretation. From the nature of the subglacial pillows in the younger material of Hengill, Saemundsson (personal communication) believes that the reversed outcrop must have been erupted under a thick glacial cover, and therefore he places its age as near the end of the Wisconsinan Ice Age, probably about 20,000 yr ago. This young age is circumstantially supported by the proximity of post-glacial volcanism about 3 km north-west of the outcrop.

We have attempted to date the reversed material radiometrically using both fission tracks and potassium/argon. No fission tracks were visible in the one glass sample which

Fig. 1 Location map, southwestern Iceland. Reversed outcrop is marked by the star south of the lake. Major normal faults are indicated by hachures, and volcanic sources are indicated by heavy lines.



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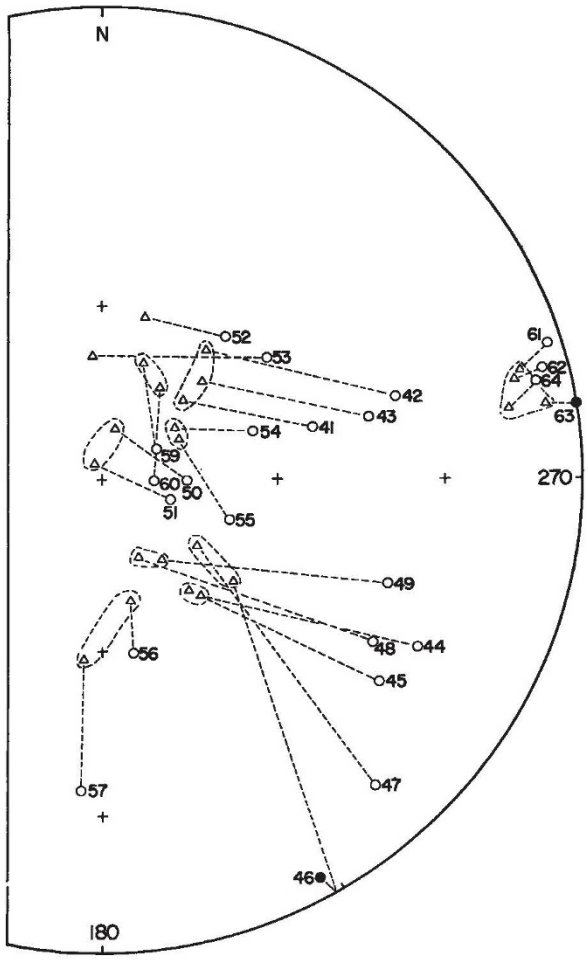


Fig. 2 Equal area plot of NRM (circles) and cleaned (triangles) directions from the individual specimens. Open symbols indicate negative inclinations. Dashed circles enclose multiple specimens from each of the 11 outcrops.

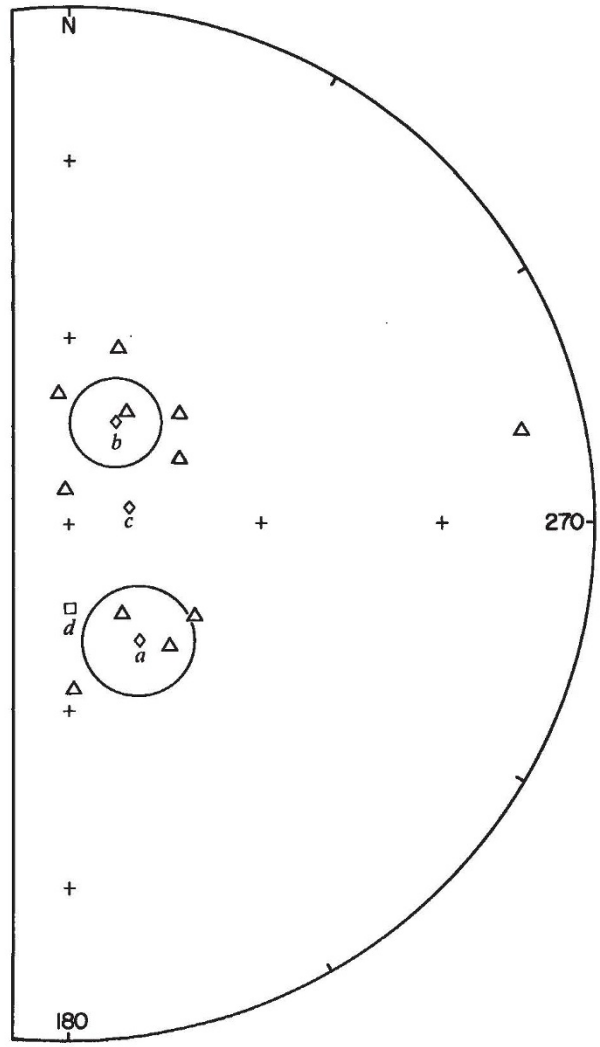
we tried to date. However, as the quality of the etching was poor, and the uranium content is very low (Gibson, personal communication), no maximum age estimate was possible. The low potassium content (0.016%) made K/Ar dating impossible. Therefore, until more suitable material for dating can be found, we can only place the age of this outcrop as upper Wisconsinan.

Until a firm age can be ascribed to these Icelandic reversed rocks, we suggest the name 'Maelifell event' for the geomagnetic event which we have inferred from them. On the basis of the stratigraphic relationships, the Maelifell event seems to be too young to be the Blake event[7]. It is more likely that this is the same as the Lake Mungo event[8,9], radiocarbon dated between 26,000 and 30,000 yr. As the Laschamp event[10,11] has been recently redated at 33,500 yr by thermoluminescence (Valladas, personal communication), we suggest that the Laschamp, Lake Mungo, and Maelifell events are the same. Clearly, further dates are needed, especially in Iceland, to confirm this suggestion. It is not clear whether the numerous reports of geomagnetic excursions in sediment cores[12,13] represent this event or whether they record some other geomagnetic variations.

If the geomagnetic field reversed recently, one would expect to find evidence of a large increase in atmospheric ^{14}C due to increased production rates during the periods of low field strength associated with polarity transitions. If the transitions were each 2,000 yr long, the increase would be about 25%. Longer transitions imply larger increases. However, a recent calibration of the radiocarbon time scale against $^{230}Th/^{234}U$ ages as old as 32,000 yr (refs 14,15) indicates no increase in ^{14}C over 8%. If this calibration is correct, then it implies either that the Maelifell event occurred at least 40,000 yr ago, or that the transitions were much more rapid than 2,000 yr, or that there was no major worldwide reduction in geomagnetic intensity associated with the event. This last possibility suggests that the Maelifell event and other synchronous events are the products of the growth and decay of an intense non-dipole feature rather than a true dipolar reversal.

In addition to supporting the concept of a young geomagnetic event, our data have major implications for the fine-scale tectonic model of the Mid-Atlantic Ridge. Although several explanations other than a geomagnetic event have been suggested to explain the axial valley reversed occurrences[2], none of these is satisfactory in Iceland. The presence of this reversed outcrop in an area which is very similar tectonically to the Mid-Atlantic Ridge suggests that similar reversed units there[1-4] cooled during this reversed geomagnetic event. However, as this event has so

Fig. 3 Equal area plot of sample directions (triangles) derived from averaging the specimen directions for each outcrop. The directions fall into two groupings whose mean directions (diamonds) and 95% confidence circles are shown. The isolated direction near 270 represents one outcrop. The direction of the reversed axial dipole ($I=-76.4$, $D=180$) is indicated by the square. The relevant statistics are: *a*, $n=4$, $R=3.947$, $k=56.6$, $\alpha_{95}=9.3°$, mean $I=-68.2°$, mean $D=210.5°$; *b*, $n=6$, $R=5.907$, $k=53.8$, $\alpha_{95}=7.8°$, mean $I=-72.1°$, mean $D=335.2°$; *c*, $n=10$, $R=9.914$, $k=15.4$, $\alpha_{95}=11.3°$, mean $I=-80.5°$, mean $D=284.1°$.



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far only been discovered at one locality in Iceland, its duration was probably quite short. But why has it been observed so frequently on the Mid-Atlantic Ridge? A few of the reported reversed samples may be due to orientation errors, perhaps in combination with extreme secular variation, but the high frequency of occurrence cannot be discounted. Perhaps this indicates that very young lavas (<50,000 yr) form a much greater part of the surface veneer of volcanism in the axial valley than previous mapping[16] has suggested. Until there is a statistically reliable estimate of the occurrences of such reversed blocks in an axial valley region, this question is debatable.

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Early Cretaceous basalt volcanism and initial continental rifting in Benue Trough, Nigeria

THE Benue Trough, located at the Gulf of Guinea re-entrant on the west coast of Africa (Fig. 1) is a sediment-filled 'failed rift', forming the third arm of a Cretaceous triple-rift (rrr) system that involved the South Atlantic and the Gulf of Guinea[1–3]. The rift nature of the Benue Trough was first documented by King[4] on geomorphological grounds, and this supposition has since been confirmed by geophysical[5], stratigraphic[6] and faunal evidence[7,8]. The considerable interest shown in the stratigraphic and tectonic evolution of the Benue graben is due to its relationship to the opening of the South Atlantic and the initiation of continental drift[9–11]. The age of the oldest Albian sediments within the trough, and their faunal biostratigraphy have been used to date the continental separation of Africa from South America[7], and infer the history of seafloor spreading in the equatorial Atlantic region[12]. New evidence is presented here on the nature of the early Cretaceous volcanism in the Benue Trough, and its petrogenesis is shown not to be due to a shortlived subduction or continental collision.

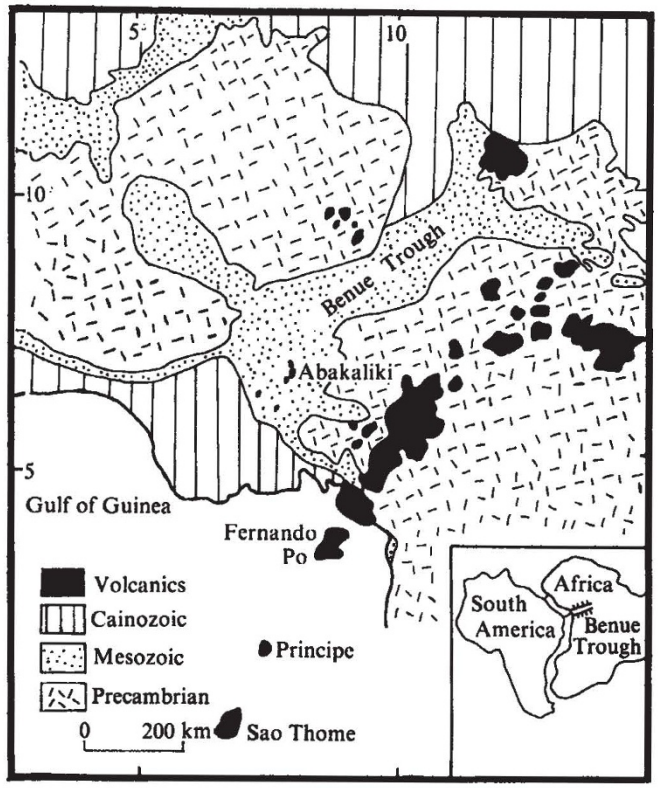


Fig. 1 Location of the Benue Trough: (inset) the Benue Trough as one arm of a Cretaceous. (rrr) triple junction.

Uzuakpunwa[13] has shown that the initiation of mid-Albian sedimentation in the Benue Trough, and by inference the opening of the South Atlantic was preceded by pre-Albian (Aptian) igneous activity involving outpouring of lavas and pyroclastic flows, probably contemporaneous with similar activity in eastern Brazil (Serral Geral lavas)[20]. These volcanic rocks ('The Abakaliki Pyroclastics'), representing eroded remnants of more extensive flood volcanics[3], are presently only well exposed at the core of anticlinal structures in the southern portion of the Benue Trough, especially around Abakaliki (Fig. 1). They are, however, older and distinct from the more widespread Upper Cretaceous to Recent alkaline rocks in the middle and upper Benue Trough. The 1,300 m of 'andesite lavas and tuffs' claimed by Burke *et al.*[4] to be interbedded with

Table 1 Selected major and trace elements in the Abakaliki volcanic rocks

Sample no.	1	2	3	4	5	6	7	8	9	10
wt %										
SiO_2	35.0	48.6	45.5	41.1	43.4	41.1	43.2	37.2	40.2	40.6
TiO_2	2.70	3.15	3.00	3.73	2.73	2.31	2.18	2.35	2.83	3.39
K_2O	0.05	1.89	0.05	0.01	0.93	0.08	0.13	0.02	0.61	0.74
Na_2O	3.70	4.40	3.90	4.10	3.60	3.70	4.60	5.00	0.13	3.10
P_2O_5	0.50	0.74	0.56	0.70	0.66	0.54	0.47	0.48	0.44	1.00
p.p.m.										
Zr	102	157	110	156	153	106	136	110	110	206
Y	18	18	18	20	19	21	19	18	19	22
Cr	207	219	230	313	123	384	169	193	265	211
Ni	159	130	147	200	145	258	140	143	205	132

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